Report: Hourly Bike Sharing Demand Forecasting

Executive Summary & Problem Statement

Accurate hourly demand forecasting is critical for operational efficiency in bike-sharing systems. Under-predicting leads to profit loss and over-predicting results in redundant fleet investments and high maintenance cost. This report presents a forecasting pipeline that predicts hourly bike rental counts (cnt), providing a reproducible modeling workflow for hourly rental-demand estimation.

1. Exploratory Data Analysis & Key Insights (based on hour.csv)

Analysis of hourly records (2011–2012) revealed strong non-linearities and temporal patterns:

- Working days follow a sharp bimodal commuter pattern peaking at **08:00** (approx. 470 rentals) and **17:00–18:00** (approx. 530 rentals). Weekends display a smooth curve peaking between **12:00 and 15:00** (about 370 rentals). (Fig. 1)
- Temperature (temp) displays the strongest positive correlation with demand (r = 0.40). Adverse weather (weathersit = 3: light rain/snow, only 3 instances for weathersit = 4: heavy rain) suppresses rental volume by roughly **45%** compared to clear conditions. (Fig. 2)
- temp and atemp exhibit a 0.99 correlation (Fig. 3); atemp was dropped to eliminate multicollinearity. casual and registered, which directly sum to cnt, were excluded.

2. Production-oriented Prototype

- Periodic features (hr, mnth) are transformed via sin/cos transformations to maintain smooth temporal continuity.
- Preprocessing (OneHotEncoder, ColumnTransformer, StandardScaler) and model execution are housed in a single scikit-learn Pipeline object.
- The pipeline relies solely on calendar signals and historical weather features (provided via hourly feature rows).

3. Model Selection & Quantitative Evaluation

HistGradientBoostingRegressor (fast execution, native handling of tabular feature and strong non-linear boundary learning) with an **Asymmetric Quantile Loss** was selected as the optimal production model over Poisson regression and standard mean-squared error GBDTs (best parameters: max_iter=300, max_leaf_nodes=31, learning_rate=0.1, loss='quantile', quantile = 0.9). Hyperparameters were tuned using 5-fold time-series cross-validation (TimeSeriesSplit) on historical data, evaluated against an unseen out-of-time test set (last 15% chronological split):

- **Historical Mean Hourly Lookup Baseline (working-day versus non-working-day):** Mean Absolute Deviation (MAD) = **91.5 bikes/hour**
- **HistGradientBoosting Model:** MAD = **49.8 bikes/hour**

The gradient boosting pipeline achieves a **45.6% reduction in mean absolute error** over historical averages, substantially reducing forecast error relative to the historical baseline. (Fig. 4, 5 & 6)

4. Error Statistics, Bias & Operational Trade-offs (Fig. 7)

The selected 90th-quantile model tends to overpredict demand on average, which may provide a conservative buffer against under-allocation. The appropriate quantile should ultimately be selected using explicit stockout and overcapacity costs:

- The proposed solution has a negative mean error of **-26.28 bikes/hr** (median error: **-13.27 bikes/hr**), indicating a slight over-allocation of bikes.
- Error standard deviation drops from **123.28 bikes/hr** (Status Quo) down to **72.21 bikes/hr** (Proposed Solution), narrowing error volatility.

Appendix: Visualizations

Figure 1: Average Hourly Demand Profile (Working Days vs. Weekends)

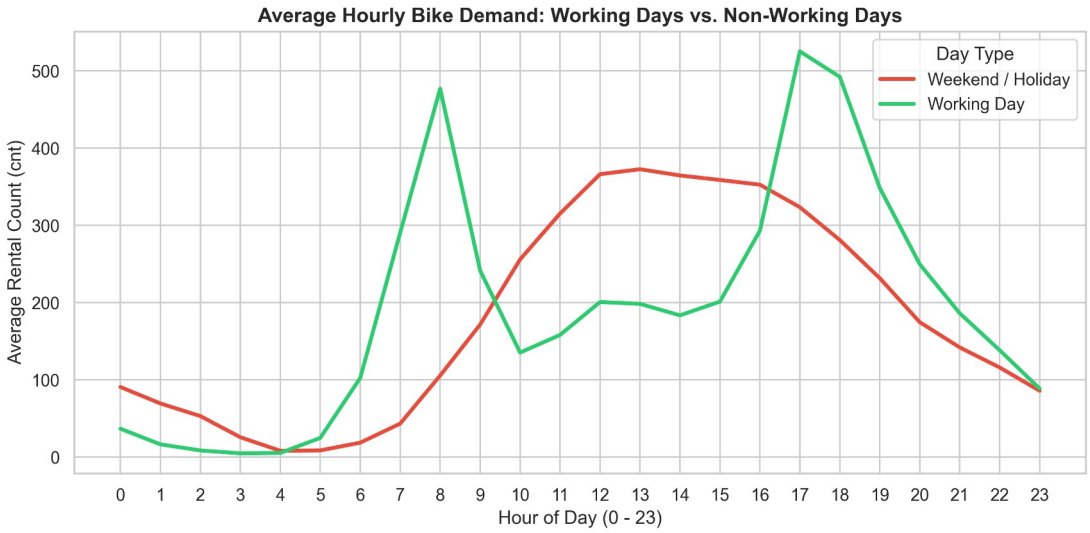


Figure 2: Weather Condition Severity & Temperature Correlation Impact

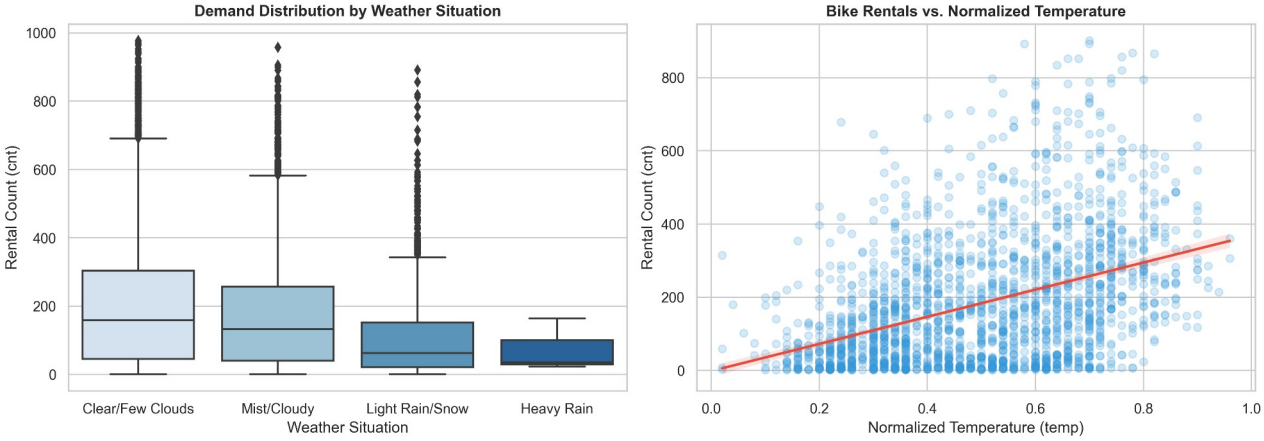


Figure 3: Feature Correlation Matrix

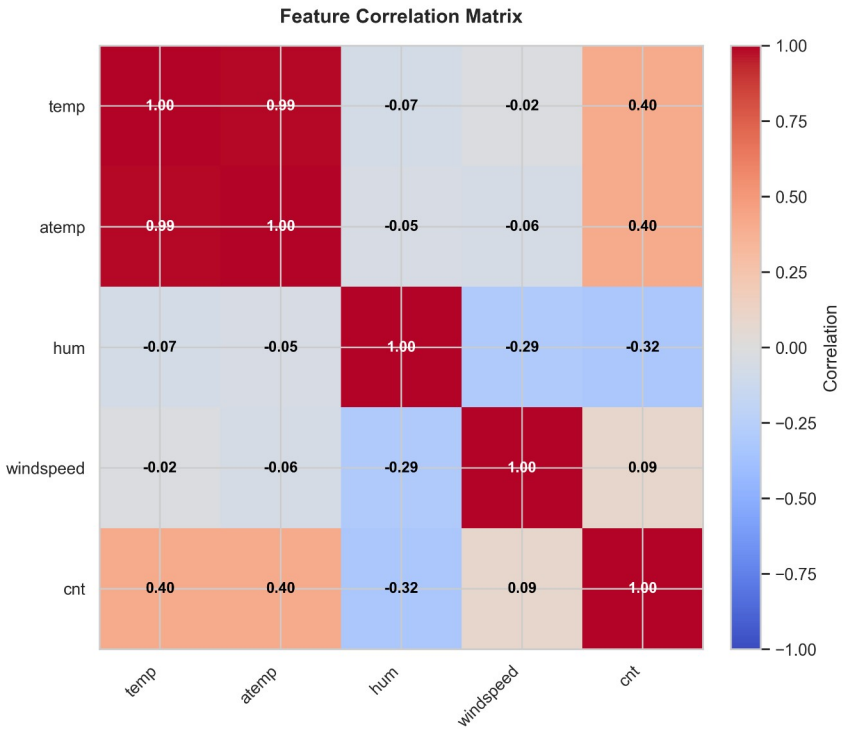


Figure 4: Actual vs. Predicted Hourly Bike Rentals

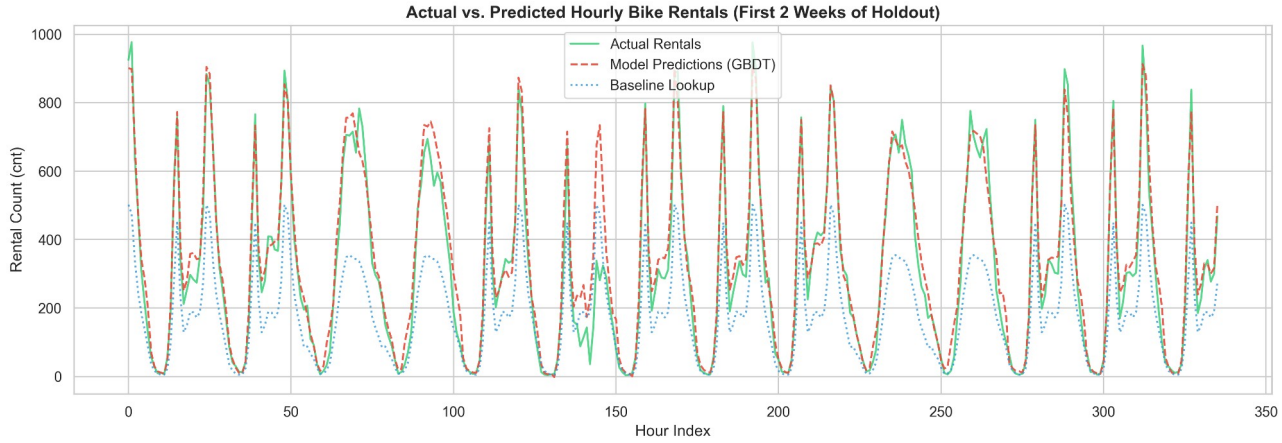


Figure 5: Forecast Accuracy Improvement

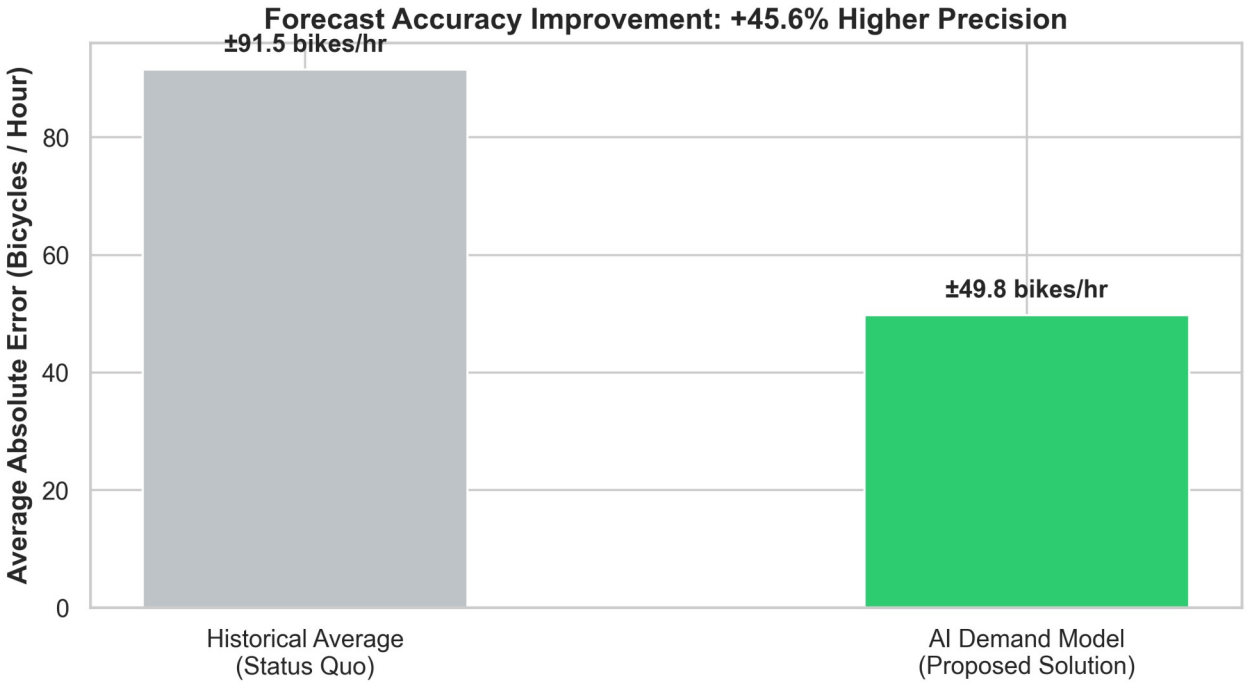


Figure 6: Baseline vs AI Model Error Distribution (MAE)

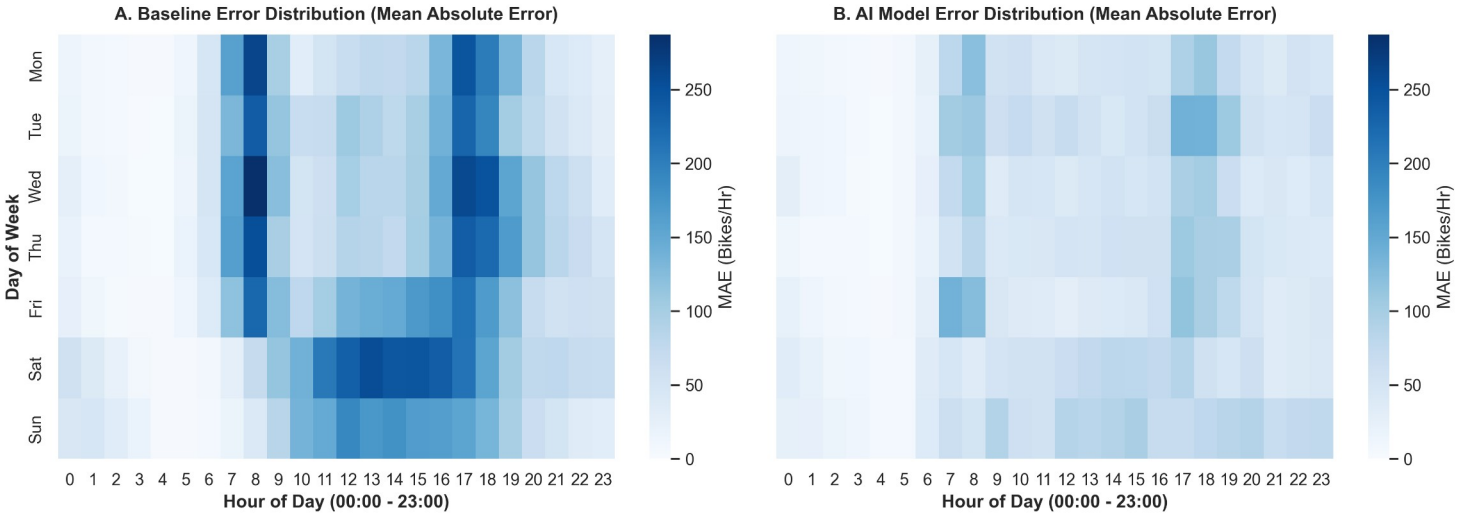


Figure 7: Error Distribution & Residual Overestimation Bias

